

Solar Home Lighting

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CONCEPT OF OPERATIONS

CONCEPT OF OPERATIONS FOR Solar Home Lighting

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1. Executive Summary

This project presents the design of a unified, solar-powered residential lighting and monitoring system intended to address growing concerns related to energy reliability, household security, and fragmented smart-home solutions. Homeowners increasingly face rising electricity costs and frequent power disturbances, while existing commercial systems often depend on multiple separate devices that operate with poor coordination relying entirely on grid power. As a result, security monitoring, lighting control, and user interaction frequently become inconsistent or unavailable during periods of power instability.

The proposed system integrates a photovoltaic charging network, battery storage, modular lighting controls, and an embedded camera with AI-based recognition into a single coordinated platform that remains operational during grid fluctuations. Lighting functions are managed through sensor-driven automation, manual controls, and mobile application inputs, while the camera subsystem provides day and night visual awareness with real-time notifications to the user. The mobile application offers remote access to system status, battery information, security alerts, and light control, enabling flexible interaction regardless of the user's location.

By combining security awareness, lighting management, and user interface capability into one platform, the system provides a reliable and self-sustaining solution suitable for modern residential settings. Its development requires interdisciplinary collaboration across hardware design, embedded firmware, renewable-energy systems, and mobile software. The result is a user-centered, energy-efficient system designed to maintain continuous operation while improving the overall safety and experience of the homeowner.

2. Introduction

This project introduces a solar-powered residential lighting and monitoring system designed to reduce dependence on the electrical grid. The system is intended to address common issues faced by homeowners, including rising energy costs, inconsistent grid reliability, and the fragmentation of current lighting and basic security solutions. By integrating sensor-based lighting control with a mobile application for system monitoring and user interaction, the design provides continuous operation and improves accessibility within a variety of residential environments. The objective of this project is to present a coordinated, renewable-energy-driven approach that enhances reliability while remaining straightforward for homeowners to operate.

2.1. Background

Residential lighting, monitoring, and security systems have traditionally relied on grid electricity and often operate as separate, uncoordinated devices. Standard lighting systems provide only basic illumination and offer limited automation, while commercial smart-security products depend on continuous electrical and network connectivity to deliver features such as live video, notifications, and remote access. During power outages or network interruptions, these systems frequently become unavailable, reducing reliability at times when consistent operation is most needed. Existing solar-powered lighting products address part of this issue but are generally limited to simple outdoor use and do not integrate with broader home functions such as indoor lighting control or coordinated monitoring.

Rising energy prices and declining grid reliability underscore the limitations of grid-dependent residential systems. According to the U.S. Energy Information Administration, the national average residential electricity price increased from 12.6 cents per kilowatt-hour in 2012 to more than 16.3 cents per kilowatt-hour in 2023—an increase of over 29%. Several states, including California, New York, and Massachusetts, have experienced increases exceeding 40% over the past decade. At the same time, the frequency and duration of power outages have grown significantly; the U.S. Department of Energy reports that weather-related outages have increased by roughly 50% since the early 2000s. These trends create growing interest in energy systems that remain functional without depending solely on the electrical grid.

Solar adoption has risen in parallel with these challenges. More than 4.3 million U.S. homes now operate with rooftop photovoltaic systems, with residential solar capacity more than doubling between 2018 and 2023. This shift reflects a broader movement toward decentralized, homeowner-controlled energy systems that offer greater resilience and cost predictability. However, most consumer-grade solar lighting products are not designed to manage multiple indoor and outdoor lighting zones, interface with security devices, or provide meaningful awareness during grid outages.

Security considerations further highlight the need for more robust, reliable residential systems. According to FBI reporting, the average financial loss per residential burglary exceeded \$97,000 in 2022, and multiple studies show that visible lighting and monitoring systems serve as effective deterrents. For example, research from the University of North

Carolina at Charlotte found that approximately 60% of surveyed offenders would avoid homes with clear signs of active alarms or surveillance equipment. Maintaining consistent lighting and monitoring capability—especially during power disruptions—therefore plays a critical role in basic home safety.

Recent advances in low-cost microcontrollers, embedded artificial intelligence, and low-power wireless communication now allow residential lighting and security functions to be integrated into a single, coordinated platform. Modern controllers can perform local sensor processing, object detection, and communication without relying on cloud services or significant power reserves. These capabilities enable a level of system autonomy and coordination not previously available in consumer-grade solar lighting systems.

Collectively, these trends demonstrate the need for a unified residential platform that reduces dependence on the electrical grid, remains operational during outages, and integrates lighting and basic monitoring functions into a coordinated, reliable system.

2.2. Overview

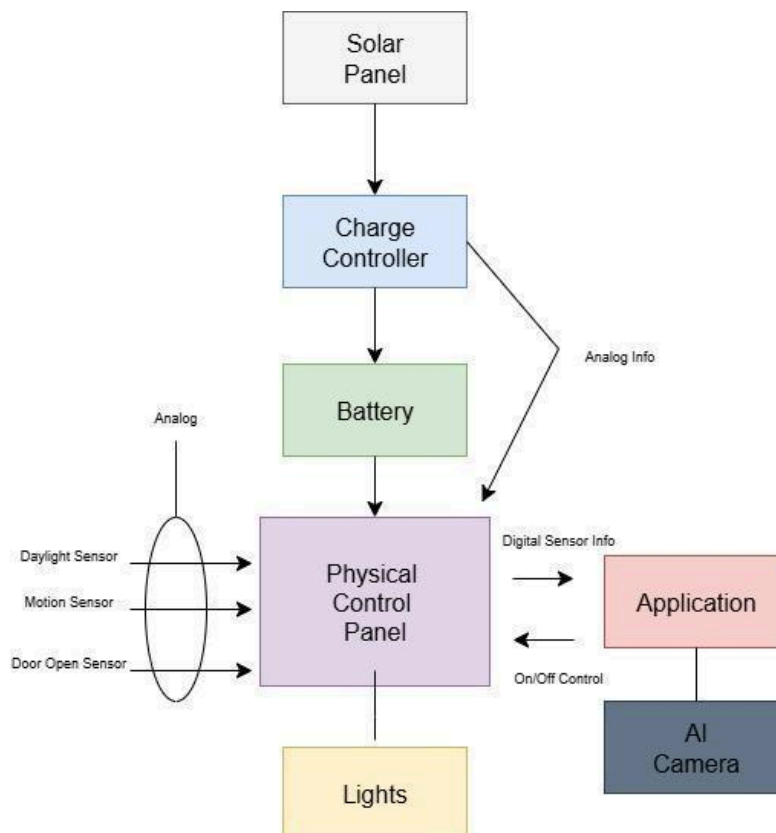


Figure 1: Solar Panel Security System Block Diagram

Our system integrates renewable energy and camera security systems by using a solar panel and charge controller to supply the solar power to a battery. This then works by powering a physical control panel built around multiple ESP32 microcontrollers. The control

panel will then take in analog signals from daylight, motion, and door opening sensors. In contrast, an ESP32 and motion sensors will also handle person detection, allowing the camera to capture human presence and send security notification signals to the mobile application. In the application, the user will be able to receive camera gallery, battery percentage, light switch controls, and security notifications. Conclusively, this design provides a robust, low-cost, energy-efficient, and user-friendly solution for a security and light system in a residential home.

2.3. Referenced Documents and Standards

- [1] IEEE Standards Association, IEEE Standard for Information Technology—Telecommunications and Information Exchange Between Systems Local and Metropolitan Area Networks—Specific Requirements Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications, IEEE Std 802.11™, 2020. Accessed at: https://standards.ieee.org/standard/802_11-2020.html

- [2] Apple Inc., Human Interface Guidelines for iOS, Apple Developer Documentation, 2023. Accessed at: <https://developer.apple.com/design/human-interface-guidelines/ios/overview/introduction/>

- [3] International Electrotechnical Commission (IEC), IEC 61724: Photovoltaic system Performance monitoring - Guidelines for measurement, data exchange and analysis, IEC Standard, 2017. Accessed at: <https://webstore.iec.ch/publication/33622>

3. Operating Concept

3.1. Scope

The project encompasses the development of a solar-powered residential lighting and monitoring system intended for use in both indoor and outdoor environments. The system is designed to operate independently of the electrical grid through the use of renewable energy storage and to provide coordinated lighting and basic security awareness. User interaction is supported through both local controls and a mobile application. For demonstration purposes, a representative model will be constructed to illustrate core functionality, while the overall design remains applicable to full-scale residential deployment.

3.2. Operational Description and Constraints

The system is intended to provide a modular lighting and security camera network for residential homes that is functioning entirely on solar power, independent from the power grid. The system will be controlled off of a local physical interface as well as a Wi-Fi-based app. The lighting aspect of the system will function with a dual purpose of recreational lighting for the interior and exterior of the home, as well as exterior security flood lights. The security camera portion will operate within the app to notify the user of detected motion outside of the home. There will be several motion sensors outside the house to detect possible trespassers and trigger both the camera functionality and security lights. For recreational lighting purposes, there will be motion sensors, door sensors, and a programmable timer to automatically engage or disengage the lights.

The following are constraints relating to the operational description:

- The solar panels need to be placed in an area that provides consistent direct sunlight during the day.
- The battery bank needs to be sized according to the number of lights to ensure adequate operating time.
- The physical control panel needs to be mounted within the range of the home's wifi to allow app functionality.
- Users are required to have a smartphone to use the security feature of this system.

3.3. System Description

The smart solar home lighting system integrates a renewable energy source, control electronics, and user-friendly software to provide a reliable and efficient home lighting and security system. The system consists of 4 subsystems: the solar panel network, microcontroller, sensors and lighting, and the application. This interconnected design ensures a sustainable and user-accessible solution for a modern home.

- **Solar Panel Network:** The solar panel array will be a roof-mounted unit fed through a charge controller into deep-cycle batteries. The batteries will collect a charge throughout the day while relaying the charge level to the user. The charge controller will be designed to implement MPPT charging to account for inclement weather that may interfere with charging conditions for maximum efficiency.

- **Microcontroller:** The microcontroller will interface with the mobile app, sensors, solar/battery network, and lighting. It reads inputs and applies the logic accordingly. It also manages comms, low power modes, enforcing limits when needed.
- **Sensors and lighting:** Motion sensors and cameras will communicate with the MCU when motion is detected. Once sensors are triggered, home lighting for designated areas will turn on, and users will be notified through the mobile app. Areas with cameras will turn on once motion is detected.
- **Application:** The Application will act as a tool that stores important system data (Camera, battery percentage, light status, and database logs) while also acting as a wireless control panel for basic operation, such as turning lights on and off. The application will communicate through Wifi via Android device and an ESP32. The ESP32 will be mounted and connected to our main control panel, where signals will be able to receive and send out vital data.

3.4. Modes of Operations

The smart Solar-powered lighting and security system supports three modes of operation - Automatic, Manual, and Security. Each is designed to adapt system behavior based on environmental conditions, user interaction, and security needs. These modes determine how the lighting subsystem, sensors, camera, and user interface cooperate to achieve the efficient and reliable operation under various scenarios.

3.4.1 Automatic Mode

Automatic mode enables the system to make lighting decisions based on real time sensor data and preconfigured user settings. It is designed to maximize energy while providing convenient adaptable control without requiring constant user input.

Operational Behavior:

- Lights turn on automatically when motion is detected under low light conditions defined by the user.
- Lighting remains active for a configurable timeout period; if no motion is detected after a set interval the lights shut off.
- Daylight conditions suppress outdoor lighting activation to preserve battery power.
- Users may adjust the motion threshold, timeout duration, and light level thresholds through the application.

3.4.2 Manual Mode

Manual mode provides full user control of the system, overriding any automated decision making. Users operate lights directly through local controls and the mobile application, enabling predictable control when automation is unwanted. This is intended to put user intent first, disabling automated behavior like sensor driven response.

Operational Behavior:

- Lights respond exclusively to user commands issued through one of two control options, physical or application based.

- Sensor inputs are ignored for any light zone placed into manual operation.
- The physical and application controls display real time system feedback, including the current light state and battery level for user consideration.

3.4.3 Security Mode

Security mode supports outdoor awareness by coordinating the lighting, camera, and AI recognition system when activity occurs around the home. This mode enhances situational awareness by providing illumination and processing visual information when motion is detected. The integrated AI model analysis video frames to determine and notify the user of situations occurring outside the home.

Operational Behavior:

- When movement is detected outside the security flood lights are triggered alongside the camera.
- The camera begins automatically capturing video that can be viewed real time through the application.
- The AI model analyzes the camera feed to identify the source of motion, and when necessary alerts the user through descriptive push notifications.
- During Daylight hours, the camera systems continue operation without the addition of flood lights to conserve system power.

3.5. Users

The smart solar home lighting system is intended to be used by any homeowner. The system is simple and user-friendly, requiring minimal technical training for standard operation. To interact with the system, the users will be provided with an in-app manual that walks them through the use of the system.

For installation, basic home improvement skills are needed to mount the solar panels and route cables to the lights. The design will contain modular sensors and light plugs that allow for ease of installation.

This setup will benefit typical homeowners who want to lower their electricity bill and desire a more vigilant mindset on home security. Additionally, this system will benefit homes in more remote areas with residences separate from the local power grid.

3.6. Support

A user manual will be provided with the system that gives details on use, installation, and troubleshooting. Additionally, the app will have a walkthrough mode available to teach users the process for controlling and monitoring the system within the application.

4. Scenarios

4.1. *Airbnb Host #1*

- Homeowners use the app to schedule porch lights to turn on at 7 PM for expected visitors.
- The system overrides motion detection during scheduled hours.

4.2. *Independent/Off-Grid #2*

- The system operates independently of grid power, ideal for cabins or remote homes.
- Reliable lighting and security without utility infrastructure.

4.3. *Nighttime Security Monitoring*

- The system is utilized when outside monitoring is needed for safety reasons
- Smart lighting and motion detectors can store camera data in the app in a safe location.
- Motion-detected lights can deter intruders.

5. Analysis

5.1. *Summary of Proposed Improvements*

The proposed solar home lighting system offers several important advantages over traditional home lighting systems, such as:

- **Eco-friendly/ Reduced Emissions:** By utilizing solar energy as the primary power source, the system reduces reliance on natural gas and lowers gas emissions.
- **Off-Grid Back-Up Use:** The integration of battery storage allows the lighting system to function independently of the electrical grid, ensuring reliable operations during outages.
- **Energy Efficiency:** The use of LED lighting and power management works to minimize energy consumption while handling high output, contributing to extending the life of the system.
- **Increased Security:** Motion sensors and app-based controls enhance home security by providing responsive lighting to movement in the area.
- **Mobile App Control:** The System can be controlled remotely via smartphone app that offers users convenience to monitor energy use, set timers, and adjust lighting preferences from any location.
- **Scalability:** The modular design allows for the expansion of a porch or foyer lighting to cover both interior and exterior household lighting, adapting to changing user needs.
- **Lower Energy Cost:** By reducing dependence on grid electricity, households experience lower monthly utility costs.

5.2. Disadvantages and Limitations

While the proposed solar home lighting system provides several advantages, there are some limitations that need to be considered, which include:

- **High Upfront Cost:** The initial investment for solar panels, batteries, and smart controls is higher than conventional lighting systems. Long-term energy savings will offset the cost, but the high upfront cost may still be a barrier for some households.
- **Weather Dependency:** Solar power generation relies on sunlight. Cloudy days, storms, or extended periods of low sunlight, such as winter, can reduce system performance, making it less reliable without sufficient battery storage.
- **Space Requirements:** Installation requires adequate roof and/or ground space for solar panels, which may not be feasible for all homes, especially in densely populated areas.
- **Energy Storage Limits:** Battery capacity restricts how much energy can be stored and used during non-sunlight hours. Once the battery is depleted, the system cannot provide power and must be supplemented with grid electricity.
- **Scalability vs Cost:** While the system can be expanded to provide lighting to more areas, expanding will incur additional cost, which could become a limitation.
- **Efficiency Loss:** Energy losses occur during solar conversion, battery charging/discharging, as well as system operations, which will reduce overall efficiency.

5.3. Alternatives

In evaluating the proposed solar home lighting system, there are several alternatives to consider, each with its own set of tradeoffs compared to the proposed system:

- **Conventional Grid-Based Lighting:** Traditional and inexpensive, but offers no sustainability or backup during outages.
- **Solar Lighting Without App Control:** A solar home system with panels, battery, and LED setup with only physical switches. This alternative is more affordable but lacks monitoring, scalability, and smart functionality.
- **Hybrid Solar System:** Combines solar with a generator or grid connection for backup. This increases reliability but adds cost, complexity.
- **Commercial Smart Home Lighting:** Uses grid electricity to power smart LEDs with wifi control. These, however, don't reduce energy cost or environmental impact.

5.4. *Impact*

- Environmental Impact: Reduces carbon emissions by using solar energy and energy-efficient LEDs. There will still be some environmental footprint through battery and panel production.
- Societal Impact: Provides reliable lighting, improves home security, and enhances user convenience.
- Ethics: May raise concerns regarding data privacy on the mobile app. Affordability for lower-income households is also a factor, as well as the sourcing of materials used in the system.